

Object Oriented Programming

CSIT — 2nd Semester — 2076 — Answer Sheet

Subject Code: N/A

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Write a program according to the specification given below:- Create a class Teacher with data members id& subject and member functions for reading and displaying data members.- Create another class Staff with data members sid& position, and member functions for reading and displaying data members.- Derive a class Coordinator from Teacher and Staff and the class must have its own data member department and member functions for reading and displaying data members.- Create two object of Coordinator class and read and display their details.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain the concept of operator overloading? List the operators that cannot be overloaded. Write programs to add two object of distance class with data members feet and inch by using member function and friend function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain types of polymorphism briefly. Write down roles of polymorphism. How can we achieve dynamic polymorphism briefly. Write down roles of polymorphism. How can we achieve dynamic polymorphism? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How object oriented programming differs from object based programming language? Discuss benefits of OOP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is the use of new and delete operators? Illustrate with example. What are advantages of new over malloc.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is meant by return by reference? How can we return values by reference by using reference variable ? Illustrate with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is destructor? Write a program to show the destructor call such that it prints the message "memory is released".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is this pointer? How can we use it for name conflict resolution? Illustrate with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. How can you define catch statement that can catch any type of exception? Illustrate the use of multiple catch statement with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is meant by template? How can you define function template to return maximum of two integers, floats, or characters? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Which functions can be used for reading and writing object? Define briefly. Write a program that reads values of two objects of student class (assume data members are sid, sname, and level) and display the data in monitor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:- Cascading of IO operators- Pure Virtual Function

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.